Stress Dector Based on Multimodal Biomarkers using Supportive Vector Machine

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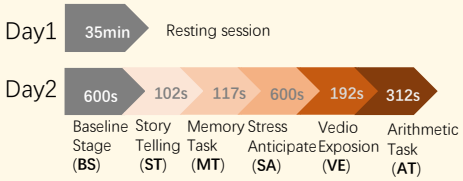
Research background

- According to studies, long-term stress can cause **mental and physical diseases** such as depression, etc.
- Stress monitoring can be achieved by analysing **biological indicators**.
- Currently, research on low-cost wearable devices primarily focuses on the **binary classification** of **stressed/unstressed** using a **single type** of physiological signal, with relatively limited attention paid to **multimodal analysis** and the **investigation of stressors**.

Research Objective

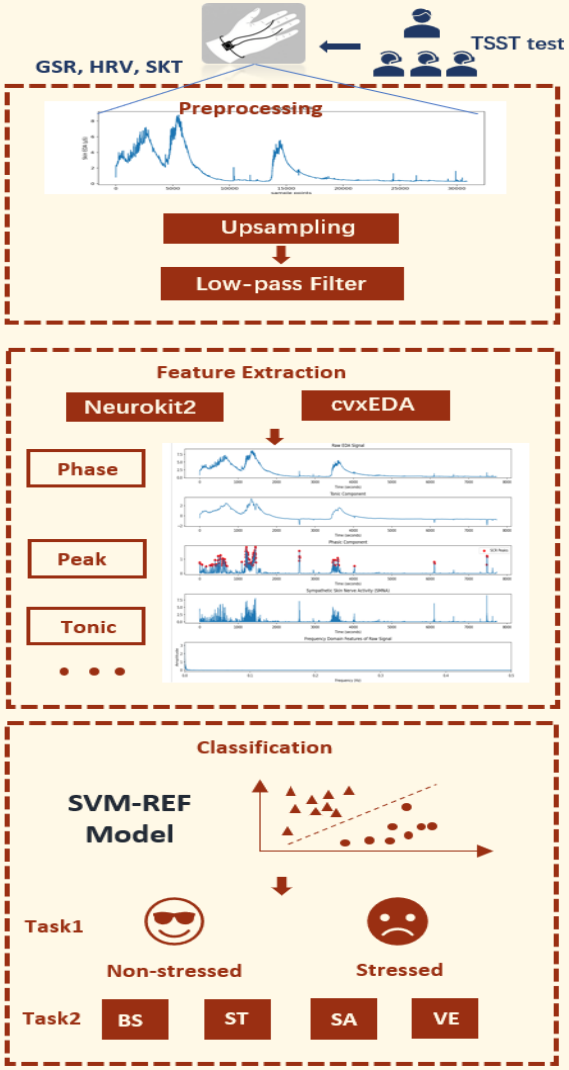
- Design a stress classifier using **Galvanic Skin Response (GSR)**, **heart rate variability(HRV)**, **skin temperature(SKT)** biosignal.
- Classify 4 different stress stages in **TSST test.**
- Build a **prototype** using available sensors

Trier Social Stress Test (TSST)



BS: help subject to relax and get ready for the stress session
ST: Subjects are required to listening and memorize 3 different strories with abundant details.
MT: Subjects are required to retell the story aloud, being recorded.
SA: The participants waited in the room for 10 minutes, during which they anticipated the evaluation results of the memory task, enhancing the stress caused by uncertainty.
VE: The video of the participants' performance in the MT stage was played, together with videos of actors correctly retelling the stories, to strengthen the social evaluative threat through social comparison.
AT: Subtract 13 continuously starting from 1022 within 5 minutes, and restart if a mistake is made, so as to increase the uncontrollability of the task.

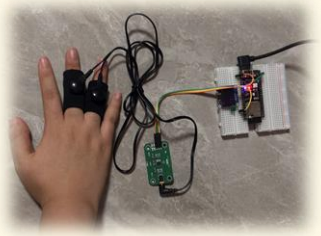
Methodology



Dataset:

Subject number	Gender: male/female	Age	Train set (in 1 epoch)	Test set (in 1 epoch)
15	11/4	27.46	14	1

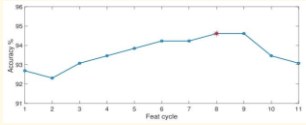
Prototype



Hardware parameter
GSR: sensor: Sichiray V2
HRV sensor: MAX30102
SKT sesnsor: MAX30205
MCU: ESP32-S3-DevkitC-N16R8

Results

Task1: Stressed & non-stressed
Accuracy trend of stress/non-stress recognition problem as a function of the number of selected features. The curve shows the accuracy considering the first N ranked features (from 1 to 11). The features are ranked according to the SVM-RFE criterion.



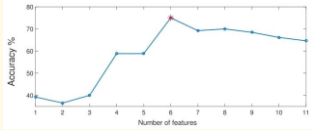
8 features: best accuracy

Confusion Matrix - Binary Classification

	Non-Stress	Stress
Non-Stress	84.62%	15.38%
Stress	2.56%	97.44%

Average accuracy: 94.62%

Task2: 4 stages in TSST test
Accuracy trend of 4-class recognition problem as a function of the number of selected features. The curve shows the accuracy considering the first N ranked features (from 1 to 11). The features are ranked according to the SVM-RFE criterion.



6 features: best accuracy

Confusion Matrix - Multiclass Classification

	BS	ST	SA	VE
BS	90.77%	6.15%	3.08%	0.00%
ST	4.62%	64.61%	4.62%	26.15%
SA	4.62%	6.15%	81.54%	7.69%
VE	4.62%	21.54%	10.77%	63.07%

Balanced accuracy: 75%

Discussion

The accuracy of the binary classification is high enough to confirm the prototype is valid. But the accuracy of the 4 class classification is under 90%, which is considered invalid. Further study should be focused on increaing 4 class classification accuracy.

References

[1] Hosseini, Elahe, et al. "A low cost eda-based stress detection using machine learning." 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). IEEE, 2022.
[2] Greco, Alberto, et al. "Acute stress state classification based on electrodermal activity modeling." IEEE Transactions on Affective Computing 14.1 (2021): 788-799.